

INTERNSHIP REPORT

2024-2025

Internship Period

17-06-2025 TO 01-07-2025 (15 Days)

SYNOVERS TECHNOLOGIES

COIMBATORE - 641009

FULL STACK DEVELOPEMENT

**COIMBATORE INSTITUTE OF ENGINEERING AND
TECHNOLOGY COLLEGE**

**DEPARTMENT OF ARTIFICIAL INTELLIGENCE
AND DATA SCIENCE**

JULY -2025

Submitted By

SIVA KUMAR S (71052308042)

INTERNSHIP REPORT
Industrial Field Training Report

SEMESTER IV

DEPARTMENT : B.TECH in ARTIFICIAL INTELLIGENCE AND DATA
 SCIENCE

COLLEGE : COIMBATORE INSTITUTE OF ENGINEERING AND
 TECHNOLOGY

COURSE TITLE : INTERNSHIP

DURATION : 17-06-2025 TO 01-07-2025

COMPANY PROFILE

NAME : SYNOVERS TECHNOLOGIES

STATUS : PRIVATE

LOCATION : NO.96C, PERANAIDU LAYOUT,
KALINGARAYAN STREET,
RAMNAGAR,GANDHIPURAM,
COIMBATORE-641009

PRODUCT : BASICS OF PHP IN WEB DEVELOPMENT

HR MANAGER : RAM KUMAR.K

WEBSITE : www.synovers.com

INTERNSHIP CERTIFICATE:



N0.96C,Peranaidu Layout, Kalingarayan Street,
Ramnagar, Gandhipuram, Coimbatore-641009

01/07/2025
Coimbatore

TO WHOMSOEVER IT MAY CONCERN

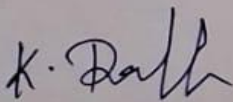
This is to certify that **Mr.SIVAKUMAR.S (REG.NO: 71052308042)**, second year **B.Tech-ARTIFICAL INTELLIGENCE AND DATA SCIENCE** student **COIMBATORE INSTITUTE OF ENGINEERING AND TECHNOLOGY, COIMBATORE** has successfully completed his Internship under **"FULL STACK DEVELOPMENT"** in our concern from **17/06/2025** to **01/07/2025**.

During this period, we found his to be sincere in his attendance. We are certifying that his conduct and performance have been found to be good.

We wish his all success in his endeavors

For Synovers Technologies

Synovers
No. 96 C, Kalingarayan Street,
Ramnagar, Coimbatore-641 009.



Ram Kumar. K
HR Manager
9994915435

ACKNOWLEDGEMENT:

I express my sincere gratitude to **SYNOVERS TECHNOLOGIES**, Coimbatore, for providing me with the opportunity to complete my internship in the field of **Web Development**. I thank the entire development team for their constant guidance, feedback, and mentorship throughout the internship.

Their expertise and willingness to help enabled me to explore advanced concepts in **frontend design, backend logic, database operations, debugging, and deployment workflows**.

I also thank **Coimbatore Institute of Engineering and Technology**, especially the faculty of the **Department of Artificial Intelligence and Data Science**, for providing continuous academic support and encouraging industrial exposure.

Finally, I thank my family and friends for their motivation and encouragement throughout this internship journey.

— **SIVA KUMAR S**

ABSTRACT:

This internship report presents a detailed account of the knowledge and practical skills gained during my 15-day Web Development Internship. The program focused on transforming theoretical concepts into real-world applications by exposing me to core web technologies such as HTML, CSS, JavaScript, PHP, MySQL, JSON, and XAMPP. Through this, I learned to design responsive interfaces, configure a local development environment, and build dynamic, data-driven web pages.

As part of my practical training, I developed two projects: a Simple Bank Web Application and an Online EB Bill Calculator. Working on these applications strengthened my understanding of both front-end and back-end development. I also gained hands-on experience in event handling, debugging, database operations, and integrating client-server communication, which helped improve my problem-solving skills.

By the end of the internship, I was able to apply the concepts learned to complete a functional mini web application. The overall experience enhanced my technical proficiency, deepened my understanding of the web development lifecycle, and prepared me for future opportunities in full-stack web development.

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Day 1:

Introduction to HTML

HTML is an application of the Standard Generalized Markup Language (SGML), which was approved as an international standard in the year 1986. SGML provides a way to encode hyper documents so they can be interchanged.

SGML is also a Meta language for formally describing document markup system. Infact HTML uses SGML to define a language that describes a WWW hyper document's structure and inter connectivity.

Following the rigors of SGML, TBL bore HTML to the world in 1990. Since then, many of us have it to be easy to use but sometimes quite limiting. These limiting factors are being addressed but the World Wide Web Consortium (aka W3c) at MIT. But HTML had to start somewhere, and its success argues that it didn't start out too badly.

Hypertext Markup Language (HTML) is the standard markup language for documents designed to be displayed in a web browser. It can be assisted by technologies such as Cascading Style Sheets (CSS) and scripting languages such as JavaScript. HTML is a computer language devised to allow website creation. These websites can then be viewed by anyone else connected to the Internet. It is relatively easy to learn, with the basics being accessible to most people in one sitting; and quite powerful in what it allows you to create. It is constantly undergoing revision and evolution to meet the demands and requirements of the growing Internet audience under the direction of the W3C, the organization charged with designing and maintaining the language.

Day 2:

Introduction to CSS

Cascading Style Sheets (CSS) is a style sheet language used for describing the presentation of a document written in a markup language like HTML. CSS is a cornerstone technology of the World Wide Web, alongside HTML and JavaScript. CSS is designed to enable the separation of presentation and content, including layout, colors, and fonts. This separation can improve content accessibility, provide more flexibility and control in the specification of presentation characteristics, enable multiple web pages to share formatting by specifying the relevant CSS in a separate .css file, and reduce complexity and repetition in the structural content.

Separation of formatting and content also makes it feasible to present the same markup page in different styles for different rendering methods, such as on-screen, in print, by voice (via speech-based browser or screen reader), and on Braille-based tactile devices. CSS also has rules for alternate formatting if the content is accessed on a mobile device. The name cascading comes from the specified priority scheme to determine which style rule applies if more than one rule matches a particular element. This cascading priority scheme is predictable.

The CSS specifications are maintained by the World Wide Web Consortium (W3C). Internet media type (MIME type) text/css is registered for use with CSS by RFC 2318 (March 1998). The W3C operates a free CSS validation service for CSS documents. In addition to HTML, other markup languages support the use of CSS including XHTML, plain XML, SVG, and XUL. CSS has a simple syntax and uses a number of English keywords to specify the names of various style properties. A style sheet consists of a list of rules. Each rule or rule-set consists of one or more selectors, and a declaration block.

Day 3:

Introduction to JAVASCRIPT

JavaScript is a dynamic computer programming language. It is lightweight and most commonly used as a part of web pages, whose implementations allow client-side script to interact with the user and make dynamic pages. It is an interpreted programming language with object-oriented capabilities. JavaScript was first known as Live-script, but Netscape changed its name to JavaScript, possibly because of the excitement being generated by Java. JavaScript made its first appearance in Netscape 2.0 in 1995 with the name Live-script. The general-purpose core of the language has been embedded in Netscape, Internet Explorer, and other web browsers. Client-side JavaScript is the most common form of the language. The script should be included in or referenced by an HTML document for the code to be interpreted by the browser. It means that a web page need not be a static HTML, but can include programs that interact with the user, control the browser, and dynamically create HTML content. The JavaScript client-side mechanism provides many advantages over traditional CGI server-side scripts. For example, you might use JavaScript to check if the user has entered a valid e-mail address in a form field. The JavaScript code is executed when the user submits the form, and only if all the entries are valid, they would be submitted to the Web Server. JavaScript can be used to trap user-initiated events such as button clicks, link navigation, and other actions that the user initiates explicitly or implicitly.

Day 4:

Xampp Setup

XAMPP is the most popular software package which is used to set up a PHP development environment for web services by providing all the required software components. During the process of software deployment, most of the web servers use almost similar components, so use of XAMPP provides easy transition from local server to live server. XAMPP is a AMP stack which stands for Cross platform, Apache, MySQL, PHP, Perl with some additional administrative software tools such as PHP MyAdmin (for database access), File-zilla FTP server, Mercury mail server and JSP Tomcat server.

Other commonly known software packages like XAMPP are WAMP, LAMP, and others.

The XAMPP server is used to test PHP pages. It works as local server. It contains a MySQL database to manage or save data on a local server.

Day 5:

Introduction to PHP

PHP stands for Hypertext Preprocessor. PHP scripts run inside Apache server or Microsoft IIS. PHP and Apache server are free. PHP code is very easy. PHP is the most used server side scripting language. PHP files contain PHP scripts and HTML. PHP files have the extension “php”, “php3”, “php4”, or “phtml”. PHP is a standard HTML file that is extended with additional features. Like a standard HTML file, PHP contains HTML tag that can be interpreted and displayed by a web browser. Anything we could normally place in an HTML file Java applets, Blinking text, server side scripts .we can place in PHP. However, PHP has three important features that make it unique.

PHP is a server-side scripting language designed for web development but also used as a general-purpose programming language. Originally created by Rasmus Lerdorf in 1994, the PHP reference implementation is now produced by The PHP Group. PHP originally stood for Personal Home Page, but it now stands for the recursive backronym PHP: Hypertext Preprocessor. PHP code may be embedded into HTML code, or it can be used in combination with various web template systems, web content management systems, and web frameworks. PHP code is usually processed by a PHP interpreter implemented as a module in the web server or as a Common Gateway Interface (CGI) executable. The web server combines the results of the interpreted and executed PHP code, which may be any type of data, including images, with the generated web page. PHP code may also be executed with a command-line interface (CLI) and can be used to implement standalone graphical applications.

The standard PHP interpreter, powered by the Zend Engine, is free software released under the PHP License. PHP has been widely ported and can be deployed on most web servers on almost every operating system and platform, free of charge. The PHP language evolved without a written formal specification or standard until 2014, leaving the canonical PHP interpreter as a de facto standard. Since 2014 work has gone on to create a formal PHP specification. During the 2010s there have been increased efforts towards standardisation and code sharing in PHP applications by projects such as PHP-FIG in the form of PSR- initiatives as well as Composer dependency manager and the Packagist repository. PHP hosts a diverse array of web frameworks requiring framework-specific knowledge, with Laravel recently emerging as a popular option by incorporating ideas made popular from other competing non-PHP web frameworks, like Ruby on Rails. PHP development began in 1995 when Rasmus Lerdorf wrote several Common Gateway Interface (CGI) programs in C, which he used to maintain his personal homepage. He extended them to work with web forms and to communicate with databases, and called this implementation "Personal Home Page/Forms Interpreter" or PHP/FI.

Day 6:

Introduction to MYSQL

MySQL Server is a powerful database management system and the user can create application that requires little or no programming. It supports GUI features and an entire programming language, Sysadmin which can be used to develop richer and more developed application. There are quite a few reasons, the first being that MySQL is a feature rich program that can handle any database related task you have. You can create places to store your data build tools that make it easy to read and modify your database contents, and ask questions of your data. MySQL is a relational database, a database that stores information about related objects. In MySQL that database means a collection of tables that hold data. It collectively stores all the other related objects such as queries, forms and reports that are used to implement function effectively. The MySQL database can act as a back end database for PHP as a front end, MySQL supports the user with its powerful database management functions. A beginner can create his/her own database very simply by some mouse clicks. Another good reason to use MySQL as back end tool is that it is a component of the overwhelmingly popular Open source software.

Is an open-source relational database management system(RDBMS) Its name is a combination of "My", the name of co-founder Michael Widenius's daughter, and "SQL", the abbreviation for Structured Query Language. The MySQL development project has made its source code available under the terms of the GNU General Public License, as well as under a variety of proprietary agreements. MySQL was owned and sponsored by a single for-profit firm, the Swedish company MySQL AB, now owned by Oracle Corporation. For proprietary use, several paid editions are available, and offer additional functionality.

MySQL is a central component of the LAMP open-source web application software stack (and other "AMP" stacks). LAMP is an acronym for "Linux, Apache, MySQL, Perl/PHP/Python". Applications that use the MySQL database include: TYPO3, MODx, Joomla, WordPress, Simple Machines Forum, phpBB, MyBB, and Drupal. MySQL is also used in many high-profile, large-scale websites, including Google, Facebook, Twitter, Flickr, and YouTube.

MySQL is the world's most popular open source database. With its proven performance, reliability and ease-of-use, MySQL has become the leading database choice for web-based applications, used by high profile web properties including Facebook, Twitter, YouTube, Yahoo! and many more. Oracle drives MySQL innovation, delivering new capabilities to power next generation web, cloud, mobile and embedded applications.

Day 7:

Introduction to JSON

JSON stands for JavaScript Object Notation. JSON is a lightweight format for storing and transporting data. JSON is often used when data is sent from a server to a web page. JSON is "self-describing" and easy to understand. The database in this case is the JSON library file, stored on the server-side and parsed upon request. As JSON was used primarily in the Trace-Me web extension to store data, I have decided it would be a good idea to use it as a method of internationalism for the Assistant of Trace-Me, mainly because of the simplicity of adding data, which is an important part of the module. By creating an array of objects, each containing default text from the Assistant and its translations, I was able to compare every text

Day 8:

PHP and MYSQL Integration

Back-End Development module unfolds the seamless integration of PHP with MySQL, a critical skill for crafting dynamic and data-driven web applications. Participants delve into the art of connecting PHP, the server-side scripting language, with MySQL, the robust relational database management system. This integration forms the backbone of back-end development, enabling the transfer of data between the web application and the database. Practical exercises immerse participants in the intricacies of establishing a secure and efficient connection, laying the groundwork for robust data manipulation. Building upon the connection, participants explore the execution of various MySQL queries, including SELECT for data retrieval, INSERT for data insertion, UPDATE for data modification, and DELETE for data deletion. This phase equips participants with the skills to interact with the database dynamically, ensuring the seamless manipulation of data based on user interactions. Real-world scenarios challenge participants to apply their knowledge, fostering a deep understanding of query execution and its implications in building functional and responsive web applications.

Day 9:

Event Handling

Today, I explored event handling in Website, a crucial skill for creating interactive and responsive apps. Event handling involves responding to user interactions, such as button clicks or gestures.

I learned how to implement event listeners in Website to detect and handle user actions. This included setting up on-click listeners for buttons and on-touch listeners for touch events.

One of the highlights of the day was building an interactive app feature. I created a button that, when clicked, displayed a toast message on the screen. This simple interaction showcased the power of event handling in providing feedback to users.

Additionally, I practiced handling more complex events, such as long presses and swipes, to expand my toolkit for creating engaging user experiences.

By the end of the day, I felt confident in my ability to implement event handling in Website apps, adding interactivity and usability to my projects.

Day 10:

Data Storage

Today's focus was on data storage in Web apps, an essential aspect for saving and retrieving user information and app data.

I learned about Shared Preferences, a lightweight and efficient way to store key-value pairs. Shared Preferences are often used for storing app settings and user preferences. Additionally, I delved into MySQL databases, which provide a structured way to store, query, and manage larger datasets within a Web app. Understanding the fundamentals of database design and MySQL queries was crucial.

To apply this knowledge, I worked on a practical exercise where I implemented Shared Preferences to save and retrieve user preferences in an app. This exercise highlighted the importance of data storage in providing a seamless user experience.

Overall, mastering data storage techniques will be invaluable when building more complex Website apps in the future.

Day 11:

Networking

Today's focus was on networking, a vital component of many web apps that require data from remote servers or API s.

I started by learning how to make HTTP requests using PHP libraries. I explored the concept of asynchronous programming to avoid blockingthe app's main thread while fetching data from the internet.

I practiced making GET and POST requests to retrieve and send data to a sample API. Understanding how to handle responses and parse data from JSON format was a crucial part of this day's learning.

Additionally, I explored libraries and best practices for handling network operations in PHP, including error handling and managing network connectivity.

By the end of the day, I had successfully implemented network features in a sample app, demonstrating my ability to fetch data from a remote server and display it in the app's interface.

Day 12:

Debugging and Testing

Today, I focused on debugging and testing Web apps, essential skills for ensuring app reliability and functionality.

I learned various debugging techniques in XAMPP, including setting breakpoints, inspecting variables, and using the debugger to step through code. These skills proved invaluable for identifying and fixing bugs in my projects.

Additionally, I explored unit testing and user interface (UI) testing in Website. Writing unit tests helps verify the correctness of individual app components, while UI tests simulate user interactions and ensure that the app behaves as expected.

I practiced writing unit tests for critical parts of my app's functionality and created UI tests to validate the app's user interface and interactions.

By the end of the day, I felt more confident in my ability to identify and resolve issues in Website apps, ensuring a smoother user experience.

Day 13:

Project Planning

Today marked the beginning of the practical application phase of my internship as I started planning for a simple Web app project. Proper project planning is crucial for delivering a successful and well-organized app.

I began by defining the app's purpose, target audience, and core features. I outlined the app's user flow, identifying the different screens and interactions required.

Creating a project timeline and milestones was another essential aspect of today's work. I broke down the project into manageable tasks and set deadlines for each. This approach will help me stay on track and organized throughout the app development process.

By the end of the day, I had a clear project plan in place, ready to start building my Website app in the coming days.

Day 14:

Web App Development

Today marked the beginning of the web app development phase, where I put my planning into action and started building the Web app project.

I began by setting up the project structure in XAMPP, including creating the necessary activities and layout files. I implemented core features and functionalities based on my project plan.

One of the significant tasks today involved creating user interfaces using HTML layout files and integrating them with PHP code. Ensuring a smooth user experience was a priority, and I worked on making the app's UI intuitive and responsive.

Throughout the day, I encountered challenges and opportunities to problem-solve, which helped me grow as a PHP developer. By the end of the day, I had made substantial progress in building my app.

Day 15: Project Completion

Today marked the successful completion of my 15-day internship, during which I finalized the web applications I had been developing. The day was dedicated to refining the projects, conducting comprehensive testing, and resolving any remaining issues to ensure smooth functionality.

I thoroughly checked each feature of both applications, fixed bugs, and enhanced the user interface for a better user experience. I also focused on optimizing performance and making improvements based on feedback received during the development process. This included streamlining workflows, validating inputs, improving loading efficiency, and ensuring that the applications behaved consistently across different user scenarios.

Simple Bank Web Application

This project simulated essential banking operations such as account creation, deposits, withdrawals, and balance inquiries. I implemented the front-end using HTML, CSS, and JavaScript to create a clean and user-friendly interface. The back-end was developed with PHP, while MySQL handled data storage securely and efficiently. Through this project, I gained hands-on experience in CRUD operations, session management, form validation, and database integration.

Online EB Bill Calculator

The second project focused on creating a utility tool that allows users to calculate their electricity bill based on unit consumption. The system included input validation, slab-wise calculations, and instant output generation. It provided a simple yet accurate solution for estimating monthly electricity costs. This project strengthened my understanding of PHP logic implementation, dynamic calculations, and error handling in user inputs.

By the end of the day, both applications were fully functional and ready for deployment. Completing these projects provided me with valuable hands-on experience and a strong sense of accomplishment. They stand as clear evidence of my progress and growth as a web developer, showcasing my ability to build complete, user-focused web applications from scratch.

CONCLUSION

This 15-day internship in Web app development has been an enriching and transformer experience. Over the course of this journey, I've acquired a deep understanding of both programming and Web app development, equipping me with valuable skills and insights. In the initial days, I laid a strong foundation in Web Development, mastering essential concepts like variables, data types, and control structures. These fundamentals have proven to be the building blocks upon which I've constructed my proficiency in Web app development. Transitioning into the world of Web development, I successfully set up XAMPP, the cornerstone of Web app creation. Navigating this robust IDE and configuring virtual devices became second nature, allowing me to focus on the creative aspects of app building.



info@synover...

to me ▾



Dear Siva Kumar S,

Greetings!

Thank you for your interest in pursuing a short-term internship in Full stack development with our organization. We are pleased to inform you that your application has been accepted for the proposed internship period from 17th June 2025 to 1st July 2025.

We appreciate your enthusiasm and are glad to offer you the opportunity to work alongside our development team. This internship will provide you with hands-on experience in Full stack development and real-time project environments.

Please find the details below:

Internship Role: Full stack development Intern

Internship Duration: 17th June 2025 to 1st July 2025

Location: Synovers Technologies, No.96c,
Peranaidu Layout, Kalingarayan Street Ramnagar,
Gandhipuram, Tamil Nadu 641009

Internship in Full Stack



Development Inbox



Siva Kumar

to info ▾



Good afternoon Sir /Mam

I'm Siva Kumar S pursuing B.Tech (Artificial Intelligence and Data Science) at Coimbatore Institute of Engineering and Technology would like to join your esteemed organization for internship. The period of internship is from 17th june till 1st of july for 15 days and I would like to do in the domain of Full stack development in your organization as your organization seems to be a perfect fit for learning.

As a part of my academic achievements and credentials relevant to the course, I would like to complete the internship as a requirement of the course in your organization. I would be grateful for the opportunity to learn from your team and contribute to any ongoing projects.

Kindly consider this requisition letter and provide an intenship. I will be looking forward to your response.

Thank you
Yours sincerely
Siva Kumar S